

Quantum Physics pset 4

1.

(a)

$$\int_{-2}^{\pi} (x^2+3) \delta(x-2) dx$$

$$x_0 = 2 \rightarrow 2^2 + 3 = \boxed{7}$$

$$\int_0^{\pi} \sin^2\left(\frac{x}{2}\right) \delta\left(x - \frac{\pi}{2}\right) dx$$

$$x_0 = \frac{\pi}{2} \rightarrow \sin^2\left(\frac{\pi}{4}\right) = \left(\frac{1}{\sqrt{2}}\right)^2 = \boxed{\frac{1}{2}}$$

$$\int_{-2}^7 \exp(x^2+4) \delta(x+3) dx$$

$$x_0 = -3 \quad -3 < -2 \Rightarrow \boxed{0}$$

$$\int_{-3}^{\infty} \frac{x^2+2x-1}{x^2+3} \delta(x+1) dx$$

$$x_0 = -1 \quad \frac{1-2-1}{1+3} = \frac{-2}{4} = \boxed{-\frac{1}{2}}$$

(b)

$$\int_{-\infty}^{\infty} f(x) \delta(cx) dx$$

$$du = c dx \rightarrow dx = \frac{1}{c} du$$

$$c > 0: \quad -\infty \text{ to } \infty$$

$$\int_{-\infty}^{\infty} f\left(\frac{u}{c}\right) \delta(u) \frac{du}{c} = \frac{1}{c} f(0)$$

$$c < 0: \quad \infty \text{ to } -\infty$$

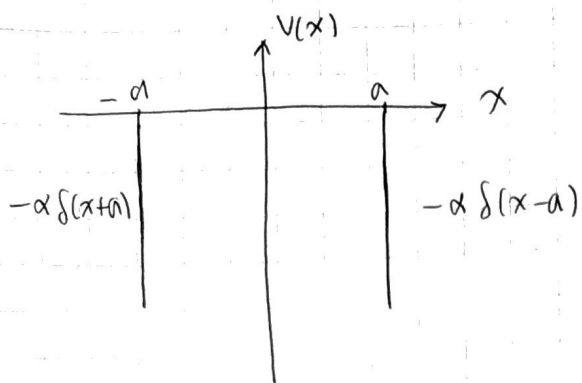
$$\int_{\infty}^{-\infty} f\left(\frac{u}{c}\right) \delta(u) \frac{du}{c} = - \int_{-\infty}^{\infty} f\left(\frac{u}{c}\right) \delta(u) \frac{du}{c} = -\frac{1}{c} f(0)$$

$$\rightarrow \boxed{\frac{1}{|c|} f(0)}$$

$$\int_{-\infty}^{\infty} f(x) \delta'(x) dx = \underbrace{f(x) \delta(x)}_{=0} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x) \delta(x) dx$$

$$= - \int_{-\infty}^{\infty} f'(x) \delta(x) dx = \boxed{-f'(0)}$$

(a)



(b)

$$k = \sqrt{-2mE/\hbar^2}$$

$$\frac{d^2\psi}{dx^2} - k^2\psi = 0$$

$x < -a$: decay as $x \rightarrow -\infty$

$$\psi = A e^{kx}$$

$-a < x < a$: no constraint

$$\psi = B e^{kx} + C e^{-kx}$$

$x > a$: decay as $x \rightarrow \infty$

$$\psi = D e^{-kx}$$

(c)

$$x = -a: A e^{ka} = B e^{ka} + C e^{-ka}$$

$$x = a: B e^{ka} + C e^{-ka} = D e^{-ka}$$

(d)

$$-\frac{\hbar^2}{2m} (\psi'(a^+) - \psi'(a^-)) - \alpha \psi(a) = 0$$

$$\psi'(a^+) - \psi'(a^-) = -\frac{2m\alpha}{\hbar^2} \psi(a)$$

$$x = -a: \psi'(-a^+) - \psi'(-a^-) = -\frac{2m\alpha}{\hbar^2} \psi(-a)$$

$$x = a: \psi'(a^-) = k C e^{ka} - k D e^{-ka}$$

$$\psi'(a^+) = -k G e^{-ka}$$

$$\text{Jump: } -k G e^{-ka} - (k C e^{ka} - k D e^{-ka}) = -\frac{2m\alpha}{\hbar^2} (C e^{ka} + D e^{-ka})$$

$$x = -a: \psi(-a^-) = kAe^{-ka}$$

$$\psi(-a^+) = kCe^{-ka} - kDe^{ka}$$

$$\text{jump: } kCe^{-ka} - kDe^{ka} - kAe^{-ka} = -\frac{2m\alpha}{\hbar^2} (Ce^{-ka} + De^{ka})$$

$$\xi \equiv \frac{k\hbar^2}{2m\alpha} \quad \beta \equiv \frac{4m\alpha a}{\hbar^2}$$

$$ka = \frac{2m\alpha a}{\hbar^2} \quad \xi = \frac{\beta}{2} \xi$$

(e)

case 1: $\psi(x) = \psi(-x)$ (even)

$$\frac{e^{kx} + e^{-kx}}{2} \rightarrow B = C$$

$$x = a: B(e^{ka} + e^{-ka}) = De^{-ka} \Rightarrow D = B e^{ka} (e^{ka} + e^{-ka})$$

$$De^{-ka} = \psi(a)$$

$$\rightarrow \psi'(a^+) - \psi'(a^-) = -\frac{1}{\xi} k \psi(a)$$

$$-kDe^{-ka} - kB(e^{ka} - e^{-ka}) = -\frac{k}{\xi} B(e^{ka} + e^{-ka})$$

$$\rightarrow \frac{D}{B} e^{-ka} + (e^{ka} - e^{-ka}) = \frac{1}{\xi} (e^{ka} + e^{-ka})$$

$$De^{-ka} = B(e^{ka} + e^{-ka})$$

$$(e^{ka} + e^{-ka}) + (e^{ka} - e^{-ka}) = \frac{1}{\xi} (e^{ka} + e^{-ka})$$

$$2e^{ka} = \frac{1}{\xi} (e^{ka} + e^{-ka})$$

$$\times \xi e^{-ka}$$

$$\downarrow 2\xi = 1 + e^{-2ka} \Rightarrow e^{-2ka} = 2\xi - 1$$

$$\text{case 2: } \psi(x) = -\psi(-x) \quad \text{odd}$$

$$C = -B \quad e^{-2ka} = 1 - 2\xi$$

$$(e^{-2ka})^2 = (2\xi - 1)^2$$

$$e^{-4ka} = (2\xi - 1)^2, \quad \beta\xi = 2ka$$

$$\Rightarrow e^{-2\beta\xi} = (2\xi - 1)^2$$

(f)